

Lecture 12: Change of Measure & Risk-neutral Pricing

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1 Radon-Nikodým Process and Change of Measure

Recall in a binomial model, we have two measures: the risk-neutral measure $\tilde{P}$ and the actual measure P . Stock price usually overperforms money market under the actual measure, however, when it comes to derivative pricing, the right measure to work with is the risk-neutral measure. One key characteristic of risk-neutral measure is that under this measure, every discounted price process is a martingale. This is not true under the actual measure. Therefore, in order to price derivatives, we need to change the measure from the actual to the risk-neutral.

The bridge that connects actual measure to risk-neutral measure is the Radon-Nikodým derivative Z :

$$Z := \frac{d\tilde{P}}{dP} > 0. \quad (1)$$

In order to price derivatives at any time $t \in [0, T]$, we need conditional expectation under $\tilde{P}$, and we have the Radon-Nikodým process to handle it:

$$Z_t = E(Z|\mathcal{F}_t) = \frac{d\tilde{P}}{dP} \Big|_{\mathcal{F}_t}. \quad (2)$$

In particular, we know from binomial model that,

$$\tilde{E}(Y|\mathcal{F}_s) = \frac{1}{Z_s} E(Z_t Y|\mathcal{F}_s), \quad (3)$$

for any $\mathcal{F}_t$ -measurable random variable Y .

Given a Itô process $\{Y_t\}_{t \in [0, T]}$ with non-zero drift, we know it is not a martingale under the current measure P . However, we can change the measure to $\tilde{P}$ so that under this new measure, $\{Y_t\}_{t \in [0, T]}$ is a martingale. This means, for any $0 \leq s < t \leq T$, we have

$$\tilde{E}(Y_t|\mathcal{F}_s) = Y_s. \quad (4)$$

In view of (3), the above equality means

$$E(Z_t Y_t|\mathcal{F}_s) = Z_s Y_s, \quad (5)$$

or equivalently, the process $\{Z_t Y_t\}$ is a martingale under the current measure P . If both Z_t and Y_t are Itô processes, then $d(Z_t Y_t)$ must have no drift term. To get this differential, we need Itô's product rule¹,

$$d(Z_t Y_t) = Z_t dY_t + Y_t dZ_t + dY_t dZ_t. \quad (6)$$

2 Discounted Stock price & Measure Change

Consider a generalized geometric Brownian motion dynamics for stock price:

$$dS(t) = \alpha(t)S(t)dt + \sigma(t)S(t)dW(t), \quad S(0) > 0. \quad (7)$$

¹This is a special case of multi-dimensional Itô formula: for function $f(x, y) = xy$, where both x and y are spatial variables.

We also consider an adapted interest rate process $R(t)$ ², the discount factor is thus given by

$$D(t) = e^{-\int_0^t R(u)du}. \quad (8)$$

Note that $D(t) = f(X(t))$ for $f(x) = e^{-x}$ and $X(t) = \int_0^t R(u)du$. Applying Itô formula we have

$$dD(t) = -R(t)D(t)dt. \quad (9)$$

Let us study the dynamics of the discounted stock process $\{D(t)S(t)\}_{t \in [0, T]}$. Using Itô's product rule we have

$$\begin{aligned} d(D(t)S(t)) &= (\alpha(t) - R(t))D(t)S(t)dt + \sigma(t)D(t)S(t)dW(t) \\ &= \sigma(t)D(t)S(t)[\Theta(t)dt + dW(t)], \quad \Theta(t) = \frac{\alpha(t) - R(t)}{\sigma(t)}. \end{aligned} \quad (10)$$

We call $\Theta(t)$ the market price of risk³. In general, $\alpha(t) \neq R(t)$, so the discounted stock price process is not a martingale under the current measure P . To turn it into a martingale (for pricing purpose), we need to change measure, so that under the new measure, there is no drift term. Effectively, we need to move to a new measure $\tilde{P}$ so that, under that measure, the drifted process

$$\tilde{W}(t) := \int_0^t \Theta(u)du + W(t), \quad (11)$$

is a Brownian motion. Then under the measure $\tilde{P}$, we have a martingale.

$$d(D(t)S(t)) = \sigma(t)D(t)S(t)d\tilde{W}(t). \quad (12)$$

The Radon-Nikodým process needed to fulfill the desired measure change turns out to be fairly simple. Consider the exponential martingale

$$Z(t) = \exp \left\{ -\int_0^t \Theta(u)dW(u) - \frac{1}{2} \int_0^t \Theta^2(u)du \right\}. \quad (13)$$

We use $\{Z(t)\}_{t \in [0, T]}$ to define a new measure $\tilde{P}$. Then the process in (11) is a standard Brownian motion⁴ under $\tilde{P}$. This new measure $\tilde{P}$ is called the risk-neutral measure.

Let us examine the discounted value process of a portfolio. We knew it is also a martingale under the risk-neutral measure in a binomial market. We show that this is still true in a geometric Brownian motion market.

Let $X(t)$ be the value of a portfolio, and $\Delta(t)$ be the number of shares of stocks held at time t . Then we have

$$\begin{aligned} dX(t) &= \Delta(t)dS(t) + R(t)(X(t) - \Delta(t)S(t))dt \\ &= \Delta(t)(\alpha(t)S(t)dt + \sigma(t)S(t)dW(t)) + R(t)(X(t) - \Delta(t)S(t))dt \\ &= R(t)X(t)dt + \Delta(t)\sigma(t)S(t)[\Theta(t)dt + dW(t)], \quad \Theta(t) = \frac{\alpha(t) - R(t)}{\sigma(t)}. \end{aligned} \quad (14)$$

Using Itô's product rule, from (9), (12) and (14) we obtain that

$$d(D(t)X(t)) = \Delta(t)\sigma(t)D(t)S(t)[\Theta(t)dt + dW(t)] = \Delta(t)d(D(t)S(t)). \quad (15)$$

Since under the risk-neutral measure $\tilde{P}$, $\{D(t)S(t)\}_{t \in [0, T]}$ is a martingale, the process $\{D(t)X(t)\}$ must also be a martingale.

²Interest rate may change over time, we choose the most general setting here. But when it comes to computation of stock option price, we choose a constant interest rate r .

³There is a similar notion in portfolio management: Sharpe ratio.

⁴To prove this fact, one need to use Lévy's characterization of Brownian motion: a martingale whose quadratic variation is equal to t . It is easy to check the second part: $(d\tilde{W}(t))^2 = dt$. To show $\{\tilde{W}(t)\}_{t \in [0, T]}$ is a martingale under $\tilde{P}$, it is equivalently to show that $\{Z(t)\tilde{W}(t)\}$ is a martingale under the old measure P , so checking the drift of $d(Z(t)\tilde{W}(t))$ will be enough.

3 Risk-neutral Pricing

Recall in a binomial market, we drive a portfolio to replicate the payoff of an European option at maturity. Then by arbitrage argument, we proved that the value of the replicating portfolio at time t is the price of the option at the same time. Suppose we are able to find such a portfolio⁵ so that an European option that pays $V(T)$ at its maturity T is fully replicated:

$$X(T) = V(T). \quad (16)$$

Then since under the risk-neutral measure $\tilde{P}$, $\{D(t)X(t)\}_{t \in [0, T]}$ is a martingale. We have that

$$D(t)X(t) = \tilde{E}(D(T)X(T)|\mathcal{F}_t) = \tilde{E}(D(T)V(T)|\mathcal{F}_t). \quad (17)$$

By no arbitrage, the price of the option at time t , $V(t)$ is given by

$$V(t) = X(t) = \frac{1}{D(t)}\tilde{E}(D(T)V(T)|\mathcal{F}_t) = \tilde{E}\left(e^{-\int_t^T R(u)du}V(T)\middle|\mathcal{F}_t\right). \quad (18)$$

Again, the price at time t is the risk-neutral expectation of the discounted payoff at maturity.

Example 1. For a geometric Brownian motion market with constant parameters: r, σ , the time t price of a call with strike K and maturity T is given by

$$\begin{aligned} V(t) &= e^{-r(T-t)}\tilde{E}((S(T) - K)^+|\mathcal{F}_t) \\ &= e^{-r(T-t)}\tilde{E}\left(S(t)\exp\left\{\sigma(\tilde{W}(T) - \tilde{W}(t)) + \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right\} - K\right)^+, \end{aligned} \quad (19)$$

where the last equality is because the Brownian increment $\tilde{W}(T) - \tilde{W}(t)$ is independent of $\mathcal{F}_t$.

4 How to Hedge

The risk-neutral pricing formula in (18) is based on the assumption that we can find a portfolio to replicate the payoff of that European option. We need to find out the right replicating strategy to do so. First note that the discounted option price process is a martingale:

$$D(t)V(t) = \tilde{E}(D(T)V(T)|\mathcal{F}_t). \quad (20)$$

Probability theory tells us, if $\{M(t)\}_{t \in [0, T]}$ is a martingale with respect to the filtration of Brownian motion $\{\tilde{W}(t)\}_{t \in [0, T]}$ under $\tilde{P}$, then there exists an adapted process $\{\Gamma(t)\}_{t \in [0, T]}$ such that,

$$M(t) = M(0) + \int_0^t \Gamma(u)d\tilde{W}(u). \quad (21)$$

Now use this theorem to martingale $\{D(t)V(t)\}$, we can also find a process $\{\Gamma(t)\}$ such that

$$d(D(t)V(t)) = \Gamma(t)d\tilde{W}(t). \quad (22)$$

Recall that, on the other hand,

$$d(D(t)X(t)) = \Delta(t)\sigma(t)D(t)S(t)d\tilde{W}(t). \quad (23)$$

Because the portfolio is replicating the option, we always have $X(t) = V(t)$. By comparing (22) with (23), we obtain that

$$\Delta(t) = \frac{\Gamma(t)}{\sigma(t)D(t)S(t)}, \quad 0 \leq t \leq T. \quad (24)$$

Formula (24) is not applicable now, since we do not know what $\Gamma(t)$ is. But we will figure it out in the future.

⁵An initial investment $X(0)$ and a trading strategy $\{\Delta(t)\}_{t \in [0, T]}$.